Achieving Universal Digital Literacy for Rural India

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ABSTRACT
In this paper, we present the experience gained in implementation of a successful Digital Literacy programmes across India. The Government has launched 'Digital India' – A programme to transform India into digitally empowered society and knowledge economy.

The need for digital literacy in a country as populous and diverse as India is critical. It is seen that the gap between limited availability of resources as against vast requirement could be addressed to a large extent by use of technology, which could provide effective ways to scale up solutions and bridge these gaps. The technology and connectivity together can make a huge difference to the socio-economic levels of a community, and ultimately, the country, since true progress comes from inclusive growth.The paper describes various innovative approaches and methodologies used to implement the programme throughout India. The lessons learnt and impact study done by third party agencies have also been highlighted.

CCS Concepts
• Applied computing~E-government

Keywords
Information & Communication Technology (ICT); ICT in Education; Digital; Digital Literacy; Aadhaar; eLearning; LMS; Training; Capacity building

1. INTRODUCTION
Digital literacy is a key component of the Government of India's vision of building an empowered society as envisaged under "Digital India" initiative. Spinoff effects of digital literacy especially in the context of rural India would address a number of socio-economic issues. 'Digital Literacy' would bring the benefits of ICT to daily lives of rural population in the major thrust areas of Healthcare, Livelihood generation and Education. The key is to have sustained efforts by harnessing collective energies, strengthening partnerships and leveraging them to pull down the divisive digital wall.

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2. DEMAND & SUPPLY ANALYSIS
As per Census of India 2011, around 68.84 % of population resides in rural India which comes to around is 0.83 billion. The number of rural household is indicated as 0.17 billion. The Census data also indicates that only around 5.2% of rural households possess a computer.As per 71st NSSO Survey on Education 2014, only around 6% of rural households and 29% urban households are having a computer. Further, 27% of Indian households (16% rural and 49% urban) had access to Internet facility in the survey year, 2014. Among persons of age group of 14-29 years in rural India, nearly 18% were able to operate a computer. Similarly, for the age group of 30-45 years, the percentage drops to 4% only. Further, for the age group of 46-60 years, the percentage is just more than 1%. This highlights that around 0.16 billion rural households are not having computers and a significant number of these households are likely to be digitally illiterate.The Ministry of Electronics and Information Technology (MeitY), Government of India had approved a scheme entitled 'IT Mass Literacy' in March 2014 which was renamed as National Digital Literacy Mission (NDLM) with a target of providing digital literacy to 1 million persons covering one person from every eligible family to enable them to use IT and related applications and participate effectively in the democratic process and enhance their livelihood. Subsequently, under Digital India, a new scheme entitled Digital Saksharta Abhiyan (DISHA) was approved in December 2014 with a target to make additional 4.25 million candidates digital literate in a period of four years.

Both the above schemes were implemented concurrently and achieved a cumulative enrolment of close to 10 million candidates; out of which around 8.2 million candidates have completed the training and of these around 5.25 million candidates have been successfully certified and targets have been achieved.

3. IMPLEMENTATION
The implementation of the NDLM and DISHA Schemes is being carried out by the CSC e-Governance Services India Ltd. (CSC-SPV) which acts as the Programme Management Unit (PMU) for both the Schemes. The physical delivery of the digital literacy training under NDLM and DISHA Schemes is being done in PPP model through various partner agencies designated as Training Partners and Training Centres, which are empanelled by CSC-SPV based on a defined criterion. After completion of training, each candidate has to appear for an Assessment carried out by a third party Certifying Agency through an online Anywhere Anytime Remotely proctored Examination. The training fee is released to the Training Partners/Centres only after the candidate is successfully certified. So far, more than 1950+ Training Partners and 1,11,000+ Training Centres have been empanelled under NDLM/DISHA to provide enrollment/training to the candidates.

3.1 Role of States and Union Territories (UT)

Each State/UT was requested to nominate an existing State/UT level agency for coordination of the Scheme in their respective State/UTs. This agency actively coordinates with CSC-SPV in implementation of the Scheme in their respective State/UTs.

3.2 Definition of digital literacy

"Digital Literacy is the ability of individuals and communities tounderstand and use digital technologies for meaningful actions within life situations".

- **Course Duration:** 20 Hours (Minimum 10 Days and Maximum 30 Days)
- **Medium of Instruction:** 22 scheduled Languages
- **Eligibility Criteria:** Any citizen who is IT illiterate
- **Training Content:** The learner would undergo training under 5 modules in order to be eligible for appearing in the term-end examination/certification:

Table 2. Training Module details of NDLM/DISHA

#	Name of the Module	Learning Hours
1	Introduction to Digital Devices	2
2	Operating Digital Devices	4
3	Introduction to the Internet	2
4	Communications using the internet	6
5	Application of the Internet	6

4. REAPING THE BENEFITS

Rural population has gained immensely from the 'Digital Literacy' programmes imparted in the current schemes. The 'Digital Literacy' has brought the benefits of ICT to daily lives of rural population in the major thrust areas of Healthcare, Livelihood generation, Education, etc.

4.1 Biometric based registration& examination

Aadhaar is a 12 digit individual identification number issued by Unique Identification Authority of India (UIDAI) on behalf of Government of India to the residents of India.

In order to uniquely identify a beneficiaries and allowing them into the system, Aadhaar was considered as the unanimous choice. Hence, under NDLM/DISHA, a mechanism for identification of the beneficiary and the linkage of beneficiary identification with Aadhaar was developed. Further, an interactive online portal www.ndlm.in was designed to register the candidates and provide the desired support to all the stakeholders. The NDLM portal has been developed to track and maintain database of all the candidates who have been trained/certified through this initiative. The portal supports various analytics and reporting mechanism to the stakeholders. It's a multimodal system developed using open source technologies and hosted on cloud platform. True auto scaling mode has been implemented so as to cater to large volumes of data and information. Aadhaar seeding (inserting) was done during beneficiary registration so that UIDs of beneficiaries are included in the www.ndlm.in portal database for enabling Aadhaar based authentication during service delivery. Beneficiaries are identified for Registration/training/examination through Aadhaar. The candidate would carry their Aadhaar card or any other Identity card to the examination centre. The portal facilitates assessment & certification through remote proctored examination.

Efforts to spread IT Literacy in the country by NGO's and other industry organization have been integrated under the NDLM/DISHA Scheme. NASSCOM Foundation has coordinated and brought all such efforts to a single platform under their umbrella. For industry candidates, the training has been imparted by these agencies through their own resources and no financial support from government was provided. The candidates who are trained through existing private partners, Industry, NGOs etc. have been considered for awarding IT literacy certificates by the recognized Certifying Agency.

4.2 CONCLUSION

Under both NDLM and DISHA, 10 million beneficiaries have been enrolled across India, out of which around 8.2 million have been trained and 5.25 million have duly certified successfully. Major learnings have been in the area of large scale deployment of the scheme all over India. The operations were controlled and monitored mainly by ICT based tool. The learnings have been in the area of technology management, project management, dealing with different stakeholders etc. Keeping in view the success of the above mentioned two schemes, the importance of digital literacy to reap the benefits of demographic advantage and spread it to the rural masses, the Govt. is considering to approve a new scheme viz. **Digital Literacy Mission** for rural India to cover Six (6) Crore additional households within the next three (3) years.

5. IMPACT ASSESSMENT

Impact assessment of NDLM & DISHA have been conducted by IIT Delhi and Council for Social Development-Delhi. Random stratified sampling technique was used. Largely telephonic surveys were conducted to get feedback from the users. A sample size of 25,000-30,000 was selected.

Report from Indian Institute of Technology-Delhi suggests that:

> 49.4% unemployed respondents felt confident to explore and apply for jobs after searching from Internet.77.1 % respondents felt happy to reach anyone they want in this whole world through computer and internet. After receiving training, the respondents felt more aware about their educational needs (80.9%).

Report from Council for Social Development-Delhi suggests that:

> 40% of the respondents stated that they are able to use the digital device on a day-to-day basis for various purposes. 63.79% of the respondents stated that the training had contributed in improving their awareness levels. 59% of the respondents stated that after attending the IT literacy training, their confidence levels have increased.

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